

Adaptive Staircase
PSY 310: Lab in Psychology
Lab Report

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Name: Anaira Poddar

Enrollment No: AU2420167

Introduction

Adaptive staircase procedure is an algorithm where the stimuli is actively adjusted on-line in response to the subject's performance. The procedure adjusts based on pre-set rules and ends when the stimuli approximate the subject's threshold, making it an efficient method at the cost of lower data availability. Adaptive staircase procedures are best for low perceptibility stimuli (Hairston, Maldijan; 2009). It is used in psychophysics to measure an individual's sensory or perceptual threshold by automatically adjusting the difficulty of a stimulus based on previous responses making it highly efficient as it does not waste time on easy and hard levels unlike traditional methods, adapts on the fly, no requirement to guess threshold prior to experiment and works well in monitoring shifts in perception (Hailing, 2025).

The adaptive staircase method calculates a difference threshold or Just Noticeable Difference, by adjusting a stimulus difference based on a participant's previous response until it meets around the target discrimination level. If the response is correct, the difficulty is increased after the correct response is made, and if the response is incorrect, the difficulty is decreased after the incorrect response is made. These points in which the tracking direction reverses are called the reversals, and the initial steps are large to quickly find the region, before they're made smaller to become more precise. The final JND is calculated by finding out the mean of the stimulus intensity values at the designated reversal points (Hailing, 2025).

Method

Participant

The study had one subject, who was a female, 43 years old. Their informed consent was taken prior to the experiment and confidentiality was assured.

Materials

The study was conducted using PsychoPy v2026.2.3. It was conducted on a laptop with a monitor of dimensions 12.5", with a resolution of 1920x1080 pixels and a refresh rate of 60Hz.

Procedure

Participants were given instructions explaining the task and required response. The participants viewed the stimulus and responded using either the left or right response key according to the tilt of the stimulus which was a cross. The direction of tilt was randomly determined with equal probability. The experiment used a 1-up/3-down staircase procedure, with initial stimulus set to 10, and levels ranging from 1-20. After an incorrect response the subsequent trial was made easier and after 3 consecutive responses the trials were made more difficult. The step size

decreases from 2 to 1 to 1 to 0.5, allowing the staircase to become increasingly precise around the participant's threshold. The experiment had 100 trials and on each trial the participants' response was recorded along with stimulus level, correct response, accuracy and reaction time.

At the end of the experiment, the recorded data was saved by Psychopy for analysis and final reversals were used to estimate the participant's perceptual threshold.

Reversal	Trial	Intensity
1	74	2
2	79	6
3	86	2
4	88	4
5	93	2
Average	3.2	

Results

The measured difference threshold is 3.2 degrees, which is the average magnitude of tilt discrimination in the last run that the participant performed correctly or incorrectly, but that this experiment had converged to and the participant's true difference threshold was oscillating around, making it a reasonable estimate.

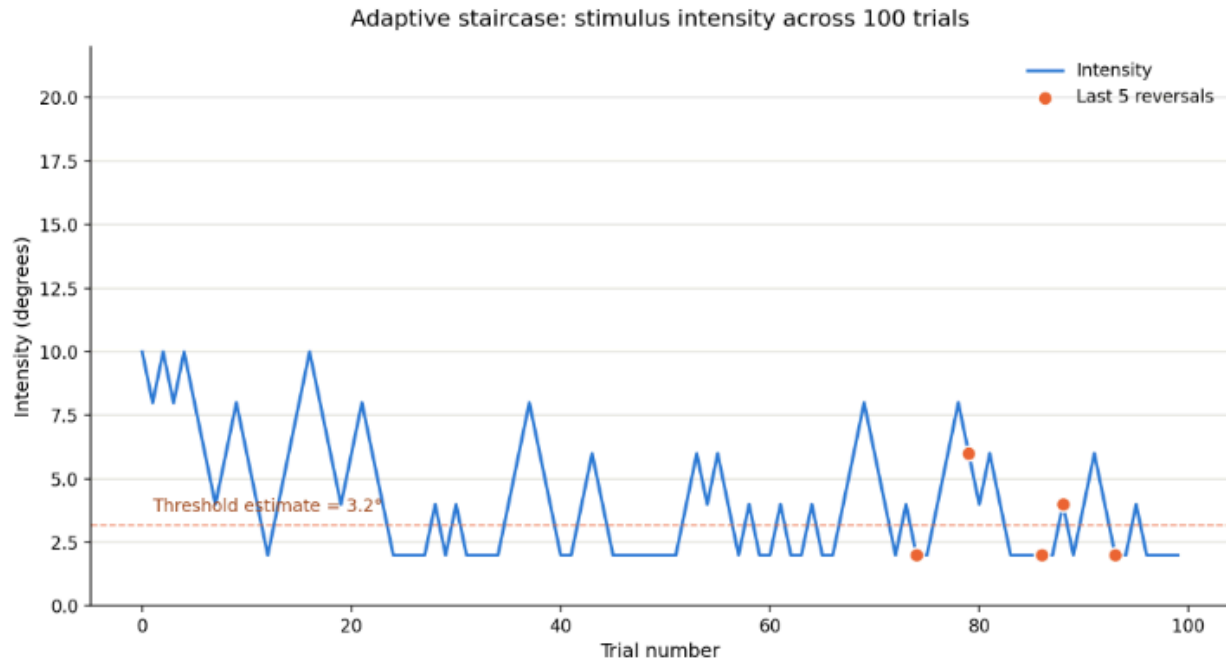


Fig 1.1: shows intensity across all 100 trials; the staircase climbs and dips repeatedly as it hones in on threshold settling mostly between 2 degrees and 6 degrees. The orange dots mark the final reversal points.

Discussion

The difference threshold obtained is 3.2 degrees, the smallest difference in orientation that the subject was able to reliably detect, and a direct measure of perceptual sensitivity to tilt. The difference threshold was low so it shows a finer discrimination ability indicating that the participant's orientation processing mechanism was able to resolve relatively small angular differences before performance broke down to chance. This threshold has been found to be within the range of 1-2 degrees of typical human orientation discrimination performance, which has been reported under a variety of conditions, including peripheral presentation, short exposure time and/or low stimulus contrast. The JND obtained is not a measure of general deficits in the participant's visual system but instead depends on the parameters of the task. While using only 5 reversals might be a bit imprecise, as the thresholds for reversals tend to converge as more reversals occur, a better estimate would be to average across more reversals or look at staircase over several runs.

References

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